

Question: Who is older, Jed Hoyer or John William Henry II?

(a) Vanilla RAG (Single-shot)

Single Retrieval Query



Jed Hoyer or John William Henry II age

Retrieved Documents (top-k)



Doc 1: "Jed Hoyer ... brother of Delta Kappa Epsilon. He married Merrill Muckerman in ..."



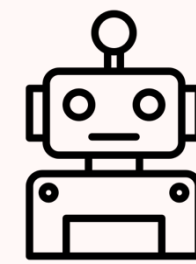
Doc 2: "John W. Henry ... net worth to be \$2.6 billion ..."



Doc 3: "Henry II of England ... born in 1133 ..."

Entity drift: retrieves "Henry II of England" instead of "John William Henry II".

Answer Generation (Single-shot)



LLM directly answers based on retrieved evidence.

Final Answer

High risk of wrong answer (due to incomplete or noisy evidence)

Limitations: No decomposition; one-shot retrieval struggles with multi-hop questions.

(b) Search Agent (Iterative)

Step 1



Think

I need to determine if Jed Hoyer or John William Henry II is older.

Search

Jed Hoyer or John William Henry II

Observe



Docs

Step 2



Think

I found Jed Hoyer is a person. I need to find his age and compare it to John William Henry II's age.

Search

John William Henry II

Observe



Docs

Step 3



Think

I found John William Henry II was born on Sep. 13, 1949. I need to find Jed Hoyer's birthyear to compare.

Search

Jed Hoyer birth year

Observe



Docs

Step T



Think

I found Jed Hoyer was born on December 7, 1993. Now I can compare birth years.

Search/Answer

Compare and answer

John William Henry II

Limitations: Unstructured trajectory; early vague queries and entity drift may lead to error accumulation and high inference cost.

(c) Ours (Decompose + Execute)

1) Decompose into subqueries

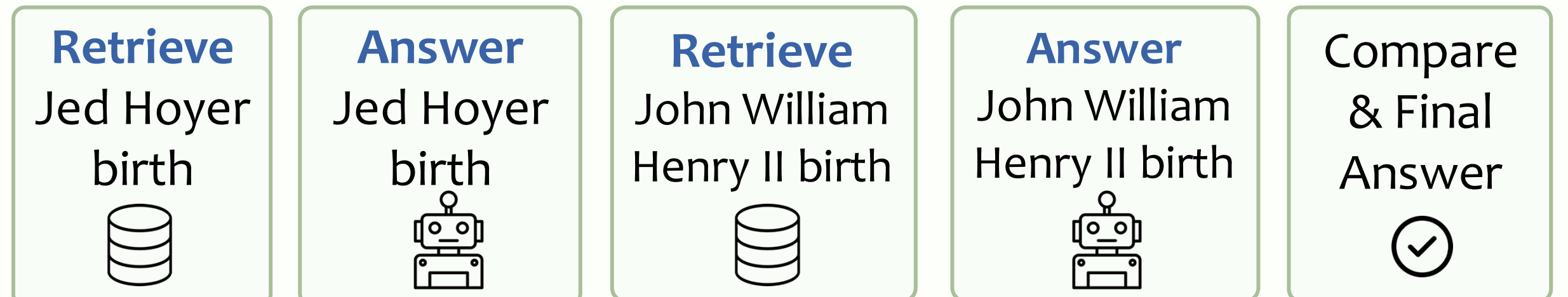
S1: When was Jed Hoyer born?

S2: When was John William Henry II born?

2) Generate Execute Plan

```
doc1 = retrieve("When was Jed Hoyer born?")
jed_birth = answer("When was Jed Hoyer born?", doc1)
doc2 = retrieve("When was John William Henry II born?")
john_birth = answer("When was John William Henry II born?", doc2)
jed_date = datetime.strptime(re.search(r"[A-Z][a-z]+\s\d{1,2}, \s\d{4}", jed_birth).group(), "%B %d, %Y")
john_date = datetime.strptime(re.search(r"[A-Z][a-z]+\s\d{1,2}, \s\d{4}", john_birth).group(), "%B %d, %Y")
return Jed Hoyer if jed_date < john_date else John William Henry II
```

3) Execute Structured Plan



Final Answer: John William Henry II

Advantages: Explicit decomposition and executable plan enable controllable, efficient, and accurate multi-hop reasoning with less drift.